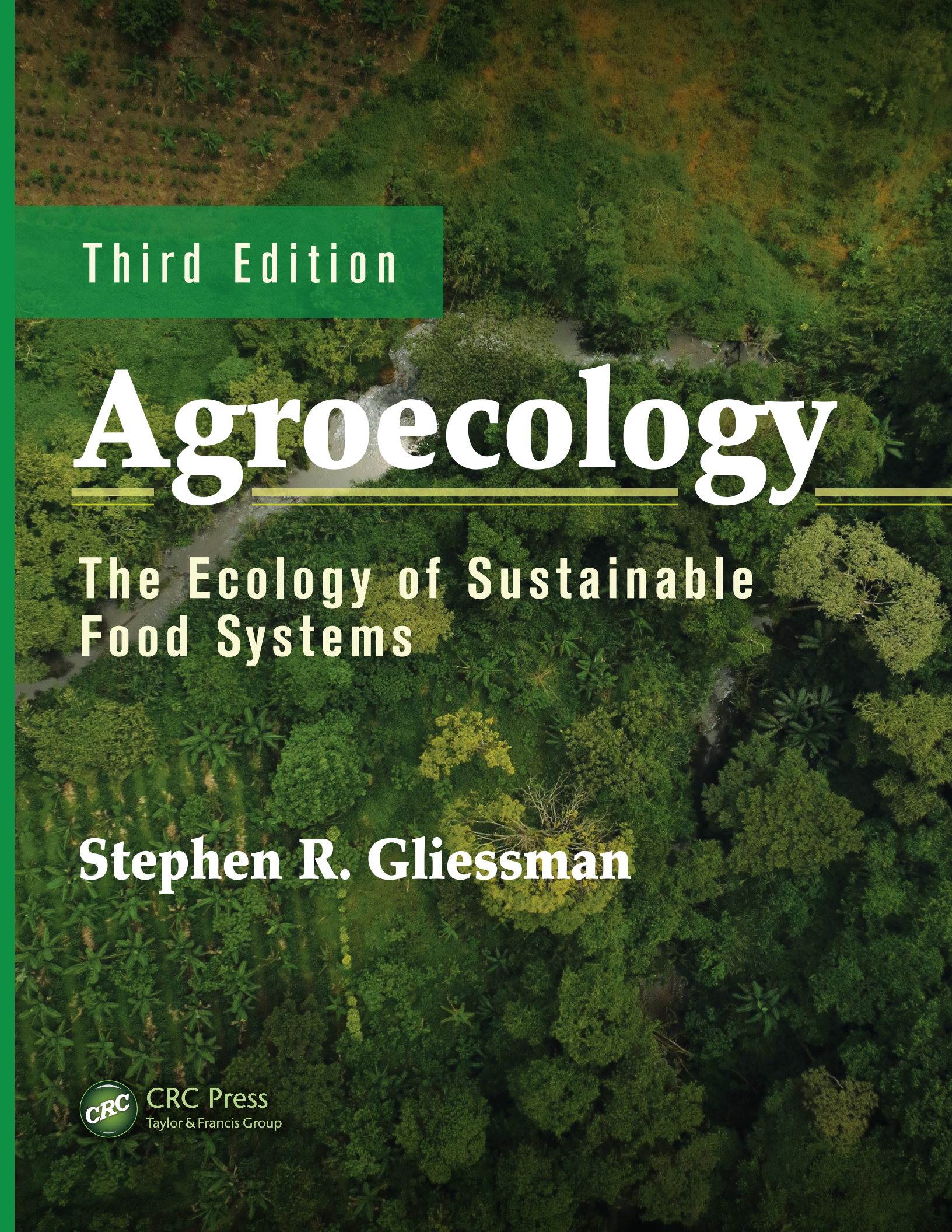

Third Edition
Agroecology
The Ecology of Sustainable Food Systems
Stephen R. Gliessman
CRC Press
Taylor & Francis Group

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Cover photo: An aerial photo of the Jimenez Family farm near AguaBuena, Coto Brus, Costa Rica, showing the integration of agriculture into a tropical forest landscape. The photo was taken during an autonomous Unmanned Aerial Vehicle (UAV) remote sensing scan. Dana Nadwodny was operating the UAV and Jonathan Dandois was assisting. They were working out of Dr. Erle Ellis' Laboratory for Anthropogenic Landscape Ecology at the University of Maryland, Baltimore, on his NSF funded Ecosynth project. The photo is one of more than 1300 frames taken to create a composite scan of the landscape in which the farm is located, and was taken on July/08/13. Photo copyright Dana Nadwodny. Used with permission. All rights reserved.

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This book is dedicated to Alf and Ruth Heller, for believing in agroecology, encouraging community, and fostering positive change at every opportunity

Contents

Foreword ix
Preface xi
Author xv
Recommendations for Using This Textbook xvii

SECTION I Introduction to Agroecology

Chapter 1 Case for Fundamental Change in Agriculture 3

Chapter 2 Agroecology and the Agroecosystem Concept 21

SECTION II Plants and Abiotic Factors of the Environment

Chapter 3 The Plant 35

Chapter 4 Light 43

Chapter 5 Temperature 55

Chapter 6 Humidity and Rainfall 67

Chapter 7 Wind 79

Chapter 8 Soil 89

Chapter 9 Water in the Soil 105

Chapter 10 Fire 117

SECTION III A More Complete Autecological Perspective

Chapter 11 Biotic Factors 131

Chapter 12 The Environmental Complex 143

Chapter 13 Heterotrophic Organisms 149

SECTION IV System-Level Interactions

Chapter 14 Population Ecology of Agroecosystems 161

Chapter 15 Genetic Resources in Agroecosystems 173

Chapter 16 Species Interactions in Crop Communities 191

Chapter 17 Agroecosystem Diversity 203

Chapter 18 Disturbance, Succession, and Agroecosystem Management 221

Chapter 19 Animals in Agroecosystems 237

Chapter 20 Energetics of Agroecosystems 253

Chapter 21 Landscape Diversity 265

SECTION V The Transition to Sustainability

Chapter 22 Converting to Ecologically Based Management 277

Chapter 23 Indicators of Sustainability 287

SECTION VI Bringing about a Sustainable World Food System

Chapter 24 Agriculture, Society, and Agroecology 301

Chapter 25 Community and Culture in the Remaking of the Food System 315

Chapter 26 From Sustainable Agroecosystems to a Sustainable Food System 335

Glossary 345

References 351

Foreword

Do not mistake this for just a textbook. You hold in your hands the key to making a difference. If life is about understanding the times in which you live and therefore what you should do with your life, then this trove of accumulated scientific insight and social wisdom from Stephen Gliessman is sure to be a key stepping stone. The reason is that Steve methodically lays out the cumulus of four decades of his experience and reflections, connecting science to purpose, action, and meaning.

With few exceptions, treatments of agriculture are about methods (how you do something) and not about the substantive questions of all human activity: what and why we do something. Tractors, fertilizers, and modified seeds are examples of some of the ways we perform agriculture. Hunger, power, and inequity are examples of attributes of agriculture and food systems and therefore why it is important to fully understand context before launching unquestioningly into methods and practices.

We are all part of a human culture (including our food system) that has settled on a predominant view of life as domination over nature and other people. Examples of the negative impacts of this domination are most obvious in the unjust conditions faced by too many people in the food system workforce, from the fields, to processing and packing plants, to shipping and stocking shelves at markets, to the food service sector. Instead of living wages, safe and healthy working conditions, and fair immigration laws that create opportunity while getting work done, we have an industrial, capital-intensive system that too often exploits both people as well as the land. We need a food system that fosters the important indicators of sustainability such as equity, fairness, and satisfaction for all, rather than domination that benefits a few. Agricultural curricula must focus beyond the narrowness of specialization, reductionism, and methodologies that primarily emphasize high yields and maximizing profits for those in power.

Are there alternatives to the current industrial food and agriculture system? That is what this book is about. If you are interested in agricultural and food systems that are managed for the long term, are more inclusive, pursue wiser purposes, and thereby converge on different methods, this book will prove an invaluable guide. It leads you through a succinct case for an alternative *understanding* of agricultural and food systems as opposed to formulaic methodologies. A farm field is not an outdoor factory, with inputs and outputs that are allegedly understood and calculated precisely on one end and maximized on the other end, scored by how much money can be squeezed from the proposition. Instead, you'll understand a farm as a cross section of many processes to be understood and integrated, where there are multiple goals, and where perpetual resilience is the key attribute because that is the way that all components do best over the long term. This includes ourselves, since, after all, humans cannot exist without supportive ecosystems.

And I should stress that this synthesis involves rigorous science, which you will enjoy immensely. One of the delights of agroecology is that it provides an answer to the very logical question for all thoughtful explorers of human knowledge: "Why should I learn this?" Here, you'll find the payoff for the basic studies you've undertaken. Physics, biochemistry, and mathematics are joined seamlessly with economics, sociology, and political science to render clearer the things that we all care about. As one example, consider a fundamental question to put to all human endeavors: "How long can we keep doing things this way?" This is the *sustainability question*, and it can be confronted competently only by combining insights from many fields of human knowledge and experience. You'll see this throughout this book, but particularly around such topics as whether we have an impending phosphorus shortage (a nonrecyclable and limited nutrient), the notion of multifunctionality and integrated landscapes, and the contextualization of the food justice movement. Gliessman is one of the world's masters of this integrated approach, and one of the best embodiments of the expertise that Robert Rodale called *metasystematics*, the discernment of how systems in perpetual motion relate with and affect one another.

This is a skill that may be unique now, but one that humanity will require in greater measure in our crowded future. As an advocate for more resilient food systems based on the principles of agroecology, I earnestly recommend that you invest yourself in developing depth and proficiency in this essential science. As with Gliessman, your credibility and effectiveness will be the more potent for your indisputable command of what Bertrand Russell called a knowledge more important than the understanding of facts, that being the connections among facts. By the time you work through this deft exposition, you will understand why agriculture must be fundamentally transformed, how ecological science can be applied to that end, and how social movements are as essential to that transformation as is the understanding of trophic interactions.

Finally, this book is a declaration of purpose: the intent to apply knowledge to improve and sustain the dignity of life for all, human and otherwise. As a book, this is an ambitious inquiry and a survey for the brave and forward-looking, from a brave and forward-looking scholar. But as an intellectual and moral challenge, this is nothing short of a call for epochal culture shift. It is time to move our knowledge about sustainable food systems beyond the

safe spaces of seminar rooms and scholarly publications. We need an *open-access* system that turns knowledge into practice, rather than into proprietary technologies owned by a few and for sale to the rest. We must understand the dynamics of economic and political power and its ability to limit, shape, and control the food system. Turning knowledge into social action on behalf of greater human well-being is the ultimate responsibility of the learned, which is to share the insights that the generosity of others has allowed one to derive, in a perpetual chain of meaning and moral action. Einstein stated, "Humanity has every reason to place the proclaimers of high moral standards and values above the discoverers of objective truth." Steve Gliessman is both a discoverer and practitioner of objective truth and a proclaimer of high moral standards. Tellingly, it is not only his many accomplished former students who can attest to this, but scores of farmers whose livelihoods have markedly improved because Gliessman is about science, purpose, and moral action. Here, he has painstakingly, joyously, fully, and generously laid out his understanding from a lifetime of effort on these fronts.

Flip the page. It is now your turn to take this on.

Ricardo J. Salvador
Director and Senior Scientist
Food and Environment Program
Union of Concerned Scientists
Washington, DC

Preface

In the late 1970s, when I and a small group of students and professors at a school of tropical agriculture in Cárdenas, Tabasco, Mexico, discussed *agroecología*—a term we thought we had invented—little did we know that agroecology would become a core part of a movement for food-system change. Much has happened in the field of agroecology since those early days, and much of that is reflected in this third edition of *Agroecology*, first published in 1996.

Agroecology has become known for being a science, a practice, and part of a social movement focused on transforming food systems to sustainability. It has also become clear how important it is that all three of these elements be integrated in a transdisciplinary, participatory, and action-oriented way in order to be most effective in bringing about the changes that are urgently needed. We now have the opportunity to move beyond the thinking of the agronomists and technologists who say that by merely increasing yields and profits we will be able to meet the food needs of the world's growing population. We were being told this when I was teaching at the Colegio Superior de Agricultura Tropical in southern Mexico in the late 1970s and the first Green Revolution was being touted as the technological miracle for agriculture. When we applied our newly emerging ecological focus to the study of these high-yield, high-input systems, it quickly became obvious that they suffered from the same problems that surround industrial agriculture today, with its focus on large-scale monocultures, huge inputs of synthetic chemical fertilizers and pesticides, and a top-down research and extension program designed to "tell farmers what to do."

Fortunately, around the college and outside the large development projects in southeastern Mexico, there was another agriculture—small-scale, traditional Maya agriculture, with 5000 years of cultural memory. For centuries, the people of the region had developed, tested, and refined practices that continue to evolve today. Featuring the traditional corn–beans–squash intercrop, whole-field *milpa* agroecosystems, the integration of crops and small livestock, complex crop associations and rotations, agroforestry, and remarkable tropical home garden systems, this traditional agriculture has fed people well for a very long time and appeared to be able to do so indefinitely. Working alongside the *campesino* farmers who managed these systems, we studied their ecological foundations, and in the process the principles of agroecology were born. At the same time, we begin to solidify our resistance to the Green Revolution model.

When I moved to UC Santa Cruz in 1981 and started the Agroecology Program, another alternative agriculture—organic farming—was just beginning to take off. Organic farming not only embodied the ecological approach we had developed in Mexico, it also served as a good foundation from which to continue developing resistance to the dominant agricultural paradigm. In the early 1980s, innovative growers were changing their farming systems to organic management, but in most cases they were doing so without much backup research to help them through the three-year transition process required for organic certification, much less help them design and manage organic crops for the long term. Through several years of collaborative farmer-based trials on their farms, we carried out transition studies in crops such as strawberries, apples, cotton, and artichokes, each with its own unique set of issues and challenges. From this experience, we adapted a system developed by Stuart Hill for redesigning food systems for sustainability as a protocol for the agroecological study of such transitional systems. We began with the first three *levels of conversion* that are described in Chapter 22. We were very successful at the outset with Level 2, where we substituted inputs and practices used in industrial systems with organically accepted ones. But as the limits of a purely substitutive approach were reached—especially when growers wanted to maintain the monoculture design they had worked with before transition—we came to the realization that a total redesign was needed to resist the problems, such as diseases, weeds, and pest insects, that came up at both Level 1 and Level 2. This became the essence for the agroecosystem redesign process that constitutes Level 3 of conversion, which was presented in the first edition of this book and is retained in this new edition.

In the years that followed the release of the first edition of *Agroecology*, it quickly became obvious that for the conversion process to reach sustainability, three levels were not enough. With only these three levels, all of the responsibility was on the farmers and all the effort was concentrated at the farm scale. We also began to observe the cooptation of organic production and markets by large growers and corporations, using their scale and market control to intensify production at Level 2. They rarely considered moving to Level 3. Organic, which in its early years was as much a philosophy as it was a way of growing crops, was being captured by the industrial food system.

I think a key milestone in our thinking in agroecology occurred with the publication in 2003 of an article entitled "Agroecology: The ecology of food systems" in the *Journal of Sustainable Agriculture*. Our team of coauthors, led by Chuck Francis, was particularly concerned with the takeover of organic agriculture, and as a response we had decided to add a Level 4 to the conversion process. This level focused on reuniting the two most important parts of the food system—those who grow the food and those who eat it. These two parts had become so isolated from one another that there was no shared knowledge among consumers about how food was

grown, by whom, or where, nor knowledge among farmers about where food went, how it was marketed, and how it was consumed. In describing this level of conversion, we were aware that alternative food networks had developed enough to become a movement resisting the dominance of the industrial food system model, with people beginning to take back their right to food system knowledge, and as my friend Rich Merrill said long ago, to put some *culture* back into agri*culture*. When the second edition of *Agroecology* appeared in 2007, the food system was a central concept, and the reconnection of growers and eaters became Level 4 in the transition process. A chapter was devoted to the alternative food system movement.

Since 2007, agroecology and our knowledge of the complexity of food system issues have grown dramatically. The global food price spikes and food riots that took place around the world in 2008 highlighted the lack of food security and access for many people in the world, which became central issues in a growing food justice movement. At the same, the rapid expansion of genetic engineering in agriculture had many extolling the promises of a "second green revolution," as corporate control of the food system became evident in everything from the seed to the market. Countering these developments, movements for food sovereignty, local and slow food, smallholder and family farms, and farmer-to-farmer organizations arose and strengthened. It became obvious to me as an agroecologist that we needed to expand the scope of the field beyond the growing and eating of food. We needed to find a political voice, align closely with social movements, and focus on developing a grassroots and community-based alternative food system that could grow outward and eventually make the industrial food system obsolete.

This commitment to social change gelled at about the same time that agroecology began finding new allies and sources of support. Important publications, such as *Agriculture at a Crossroads* (published by IAASTD in 2009) and *Agroecology and the Right to Food* (published by the United Nations Special Rapporteur in 2011), proposed agroecology as an alternative approach for resolving the interrelated global problems of hunger, rural poverty, and sustainable development. New agroecology degree programs appeared around the world: at Florida International University in Miami; Universidad de Córdoba in Andalusia, Spain; Universidad de Antioquia in Medellín, Colombia; at several universities in Brazil; and elsewhere—most with a focus that crosses the normal boundaries between natural and social sciences. Farmer organizations such as Via Campesina put forward agroecology as a primary means of creating food sovereignty, opportunity, and justice in farming communities. The science of agroecology found new outlets for its research when the journal publisher agreed to change the name of the *Journal of Sustainable Agriculture* to *Agroecology and Sustainable Food Systems* beginning in 2013. It was clear to me that we were witnessing the beginning of a paradigm shift with the potential to move the entire food system to a sustainable basis. This would entail fundamental changes in our social, cultural, and economic systems and institutions—which would go beyond what we had described for Level 4. A fifth level of conversion was needed to complete the transformation of food systems to sustainability, and thus Level 5 features prominently in the final section of this third edition.

As I look back on the 40-plus-year journey I have had in agroecology, I find two personal projects that illustrate how the creation of Level 5 came about. The first began as a group effort among some of my graduate students (listed below), my wife Robbie Jaffe, and me through our work in the coffee-growing communities of Mexico and Central America, which at the time were undergoing the worst price crash in the history of coffee as a commodity crop. We formed the nonprofit network Community Agroecology Network (CAN; described in more detail in Chapter 25), and began working with the growers at Levels 3 and 4 simultaneously. As the network grew to include non-coffee-growing communities, we quickly jumped to Level 5 with programs in food security, food sovereignty, health and nutrition, and youth leadership. Social change became linked with sustainable farming.

The other project was applying the practice of agroecology on our own farm, Condor's Hope Ranch. By combining agroecology with the traditional dry-farming practice, described in Chapter 6, for wine grapes and olives, our family developed a farming operation that we hope to someday pass on to our children, nieces, nephews, and grandchildren. In the 20 years since beginning the farm, we have dealt with multiple farming practice challenges, but the bigger issues are how a small family operation can compete in the highly corporatized industries of wine and olive oil and how we can encourage our future generations to carry on with the same passion and opportunity that Robbie and I have had. These are Level 5 issues.

Agroecology has matured as a science, is recognized as an important practice, and has aligned with a growing social movement for food system change. Our goal is to develop food systems that meet local and regional food, feed, and fiber needs, conserve and protect natural resources, provide essential environmental services, ensure food security and sovereignty, make food justice a reality, and create the opportunity for present and future generations to enjoy healthy and satisfying lives.

That may sound like a lot for the field of agroecology to take on. But I think you will see how this can happen as you make your way through the book. It begins with a strong ecological foundation for farming practices and ends with all of us thinking about the critical importance of transitioning to a new paradigm for food and agriculture, and what this means for our future.

I conclude this preface by acknowledging and thanking the innumerable people who have helped me form my agroecological vision over these many years. Among those that I value as colleagues are Itziar Aguirre, Miguel Altieri, Francisco Roberto Caporal, José Antonio Costabeber, Joao Carlos Costa Gomes, Bruce Fergeson, Chuck Francis, Roberto Garcia Espinosa, Alba Gonzalez Jácome, Manuel Gonzalez

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I deeply appreciate all that I have learned from a very special group of graduate students who have all truly earned the title of agroecologist: Jan Allison, Nick Babin, Marcus Buchanan, Rose Cohen, Wes Colvin, Ariane de Bremond, Francisco Espinosa, Michelle Glowa, Carlos Guadarrama, Kathy Hilimire, Eric Holt-Gimenez, Robbie Jaffe, Rob Kluson, Leslie Linn, Hillary Melcarek, V. Ernesto Méndez, Carlo Moreno, Joanna Ory, Jim Paulus, Francisco Rosado May, Martha Rosemeyer, Devon Sampson, and Laura Trujillo. Over the years, I have been able to collaborate with a remarkably diverse group of postdoctoral researchers: Gianumberto Accinelli, Belén Cotes, Erle Ellis, Manolis Kabourakis, Rie Mayaura, Eleonora Morganti, Joji Muramoto, Sunita Rao, Jesus Juan Rosales, Anastasia Scotto, Tatiana Sevilla, Koos Steyn, and Roberto Tinoco. I am also deeply indebted to what seems like a multitude of undergraduate students who were the initial stimulus for this textbook. Their passion for food system change and hope for the future helped transform this book into what it has become.

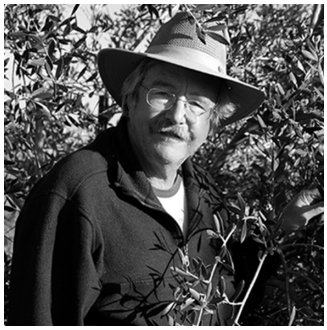
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Without a doubt, the person I owe the most for bringing this edition into existence is master editor Eric Engles. The *with* before his name on the title page has a remarkable story behind it. His capacity for keeping track of details, editing my writing, and shaping ideas, and his insistence that the full story of power and concentration be told, made this book what it is and helped greatly in adding Level 5 to the transformation process that needs to happen for the future of food, agriculture, and our planet.

Finally, I have an accumulated debt of gratitude that I owe my *compañera* Robbie Jaffe, who has supported this book project from its initial inception in the mid-1990s. Over the past year, while this edition has taken shape, she has patiently (and sometimes not so patiently) given me the space and time I needed. Be it as a skilled environmental educator, as founding executive director of CAN, or as co-farmer at Condor's Hope, her own commitment to food system change is unsurpassed. As she has shared agroecology with me, we both continue today to share agroecology with our mutual families of four generations, from which our heritage comes and to whom we pass it on.

Steve Gliessman
Condor's Hope Ranch
New Cuyama, California

Author



Stephen R. Gliessman earned graduate degrees in botany, biology, and plant ecology from the University of California, Santa Barbara, and has accumulated more than 40 years of teaching, research, and production experience in the field of agroecology. His international experiences in tropical and temperate agriculture, small-farm and large-farm systems, traditional and conventional farm management, hands-on and academic activities, nonprofit and business employment, and organic and synthetic chemical farming approaches have provided a unique combination of experiences and perspectives to incorporate into this book. He was the founding director of the University of California, Santa Cruz, Agroecology Program, one of the first formal agroecology programs in the world, and was the Alfred and Ruth Heller Professor of Agroecology in the Department of Environmental Studies at UCSC until his retirement in 2012. He is the cofounder of the nonprofit Community Agroecology Network (CAN) and is currently the president of its board of directors. He is the editor of the international journal *Agroecology and Sustainable Food Systems*. He dry-farms organic wine grapes and olives with his wife Robbie, son Erin, and daughter-in-law Oriana in northern Santa Barbara County, California.

Recommendations for Using This Textbook

Reflecting agroecology's origins in both the pure-science field of ecology and the applied field of agronomy, this text has a dual identity: In one sense, it is designed to teach ecology in the context of agriculture; in another sense, it teaches about agriculture from an ecological perspective.

Despite its attention to the practice of growing food, however, this is not a book on how to farm. Farming is an activity that must be adapted to the particular conditions of each region of the world, and this text's mission is to create an understanding of concepts that are of universal applicability.

The text has been written to accommodate a range of experience and knowledge levels in both ecology and agriculture. Sections I, II, and III assume only a basic knowledge of ecology and biology, and even those students with minimal college-level science training should have little difficulty comprehending the material if they are diligent. Intensive study of Chapters 1 through 13 will prepare any student for the more complex chapters of Sections IV, V, and VI.

Readers with extensive background in ecology will benefit most from the three latter sections. They may want to skim Chapter 2 for review, and then read Chapters 3 through 13 selectively before turning their attention to the next three sections. Readers with advanced training in both ecology and agriculture, including advanced undergraduates, may want to pursue this strategy as well, supplementing the text with additional materials that provide more extensive literature review and reports on research findings.

The text can be used in either a one-quarter or one-semester course, but the rate at which material is covered will depend greatly on the instructor, the students, and the curriculum. Ideally, a laboratory section will complement the lecture section of any course using this textbook, allowing the testing of ecological concepts in agriculture, and the demonstration of how the tools of ecology can be applied to the study of agroecosystems. The accompanying lab manual, *Field and Laboratory Investigations in Agroecology*, is designed to fill this role. Its investigations are keyed to the chapters in this text, and the two work together to create an integrated course.

Suggested readings and a list of Internet resources at the end of each chapter provide further materials for the curious reader. The questions following each chapter are open ended, designed to encourage the reader to consider the ideas and concepts presented in the broader context of sustainability.

The concepts and principles in this text can be applied to agroecosystems anywhere in the world. Just as a farmer must adjust to local and changing conditions, readers of this book are challenged to make the necessary adaptations to apply its contents to their own situations—finding appropriate examples and case studies in the research literature and working with local farmers to connect principles to actual practices.

Section I

Introduction to Agroecology

As the science of connections among living things, ecology affords a way of looking at agriculture that immediately expands its scope well beyond tilling, sowing, cultivating, harvesting, and marketing. In *agroecology*, we move from a narrow concern with farming practices to the whole universe of interactions among crop plants, soil, soil organisms, insects, insect enemies, environmental conditions, and management actions and beyond that to the effects of farming systems on surrounding natural ecosystems. Expanding this to a global scale, we see agriculture as the most land-intensive human activity on the earth, which leads us to consider the overall effects of farming on the ability of the earth to support its populations of humans and other living things. Examining human beings as a particular population, the ecological perspective then encourages us to look into the social world, at such topics as human food consumption patterns, the proportion of farmers to consumers, and the unequal distribution of food.

Casting the net of relevance this broadly leads, we hope, to an integrated perspective in which agriculture can be seen as a key factor—perhaps *the* key factor—in an intensifying crisis confronting humankind. Agriculture is not only a major cause of this crisis; it is also an arena full of potential solutions. The most basic goal of this section is to introduce readers to this greatly expanded way of thinking about agriculture.

Chapter 1 describes the many harms to people, soil, resources, and ecosystems brought about by the way we produce food today and discusses how applying ecological concepts and principles to the design and management of systems of food production—the essence of agroecology—can help us produce food more sustainably. In this way, the chapter constructs an overall context for everything we will consider in this text. Chapter 2 then outlines the fundamental concepts, theories, and perspectives that make up the framework of agroecology—thus establishing the foundation of the approach to growing food that we present in Chapter 1 as the alternative to the unsustainable system we have now. With an understanding of the stakes involved in how we humans grow our food and knowledge of the agroecosystem concept, the reader is prepared to explore the many layers of understanding that make up agroecology.



FIGURE S.1 An intensive vegetable-based agroecosystem on the urban fringe of Shanghai, China. In systems such as this, food is produced for local markets without much of the fertilizer, pesticides, and machinery characteristic of large-scale, single-crop agroecosystems.

1 Case for Fundamental Change in Agriculture

According to a variety of measures, agriculture, considered on a global scale, posted a long streak of extraordinary successes beginning shortly after World War II. During the latter half of the twentieth century, yields per hectare of staple crops such as wheat and rice increased dramatically, food prices declined, the rate of increase in food production generally exceeded the rate of population growth, and chronic hunger diminished. This boost in food production was due mainly to scientific advances and technological innovations, including the development of new plant varieties, the use of fertilizers and pesticides, and the growth of extensive infrastructures for irrigation, all of which contributed to the development of what we will call *industrial agriculture*.

Although agriculture on a global scale has more recently struggled to maintain the ever-improving trends for yield increases, food price reductions, and hunger diminishment that it achieved in the twentieth century, it remains extraordinarily productive, providing abundant food for a large proportion of the world's people. Because industrial agricultural has done a superb job of "delivering the goods," many people in the developed and developing worlds have come to take food for granted. When supermarket shelves are always stocked with a cornucopia of edible products, people tend not to devote a great deal of thought to what it takes to get the food onto the shelves. In historical perspective, this is really an unprecedented situation. Ever since *Homo sapiens* arose some hundreds of thousands of years ago, most humans have had to put the source of their next meals at the top of their list of concerns. But while having a relative abundance of food is a good thing compared to its opposite, it has tended to desensitize us to food issues, to make those of us with good access to food uncritical about how food comes to be.

Ironically, this is precisely the time in our species' history when we need to be taking stock of our food system with a more critical eye than ever before. Just because industrial agriculture is able to create food abundance in the present does not mean it will be able to do so over the long term. Indeed, it is time we came to the realization that industrial agriculture's productivity comes at a steep price and that the bill is eventually going to come due. To create the food productivity that we take for granted today, the industrial system of food production is sacrificing the basic foundations of agriculture—fertile soil, available moisture, amenable climate, nutrient recycling, genetic diversity, and the ecosystem services of natural systems. These prerequisites of food production can take only so much abuse before they begin to fail, putting at risk the food supply of tomorrow.

Another way of describing the situation is that the industrial agriculture model that dominates agriculture today is at the core of a fundamental contradiction: the techniques, innovations, practices, and policies that constitute industrial agriculture, and which have played the largest role in increasing agricultural productivity, have also undermined the basis for that productivity. They have overdrawn and degraded the natural resources upon which agriculture depends. They have created a dependence on nonrenewable, increasingly costly fossil fuels, the use of which exacerbates climate change. And they have helped to forge a system that concentrates ownership of food-system infrastructure in the hands of a few while taking it away from farmers and farmworkers, those who are in the best position to be stewards of agricultural land. In short, the contradictions inherent in our industrial agriculture-dominated system of food production make it unsustainable—it cannot continue to produce enough food for the growing global population over the long term because it deteriorates the conditions that make agriculture possible.

At the same time, our world food system faces threats not entirely of its own making, most notably the emergence of new agricultural diseases, rising costs for all the physical factors of production (land, water, energy, inputs), and climate change. As currently configured, the global food system is terribly ill equipped to face these threats. Increasingly, experts are raising red flags about the ability of agriculture worldwide to adapt to an earth on which droughts, heat waves, and extreme weather events become commonplace and the entire biosphere undergoes major shifts with potentially severe consequences for the growing of food.

Although how we feed ourselves is among humankind's weightiest issues, there is a conspicuous lack of consensus on the current status of the world food system and its future sustainability. A large number of experts—policy analysts, economists, scientists, researchers, and even some business leaders—agree with the rough outlines of the view just presented (e.g., IAASTD 2009; IFAD 2013). They believe that the industrial methods that dominate the world food system today are causing great harm to people and to earth's life-support systems and cannot (and should not) be sustained. But as numerous and authoritative as they are, these voices of concern are drowned out by those who predict productivity increases into the distant future and advocate for intensification and further dissemination of the very same methods and technologies singled out by critics of industrial agriculture as being most harmful.

The causes of this crucial difference of opinion will be addressed in the final section of this book (Section VI).

In the meantime, we encourage readers to entertain the critical perspective with which this chapter began and be open to the possibility that the world food system, as productive as it is, does in fact undermine the foundations of food production and needs to be replaced by something fundamentally different.

The first step in this direction is to take a broad and critical look at the practices of present-day agriculture—that is, to examine the largely hidden costs associated with the remarkable yields we have been extracting from the world's agricultural lands.

PRACTICES OF INDUSTRIAL AGRICULTURE

Present-day agriculture is built around two related goals: the maximization of production and the maximization of profit. These goals give agriculture a striking resemblance to the manufacturing processes that occur in factories. In both cases, elements of production are reduced to their simplest forms, processes are mechanized so that they can brought under the full control of human operators, and efficiency of output in relation to input crowds out any other goals. Although this form of agriculture is often called *conventional* to distinguish it from so-called *organic* agriculture, its factory-like nature suggests the more descriptive term used in the introduction to this chapter: *industrial agriculture*.

In pursuit of maximum production and profit, a host of practices have been developed in industrial agriculture without regard for their direct social and environmental costs or their unintended, long-term consequences. Seven basic practices—intensive tillage, monoculture, irrigation, application of inorganic fertilizer, chemical pest control, genetic manipulation of domesticated plants and animals, and "factory farming" of animals—form the backbone of modern industrial agriculture. Each is used for its individual contribution to productivity, but as a whole the practices form a system in which each depends on the others and reinforces the necessity of using all in concert.

Intensive Tillage

Industrial agriculture has long been based on the practice of cultivating the soil completely, deeply, and regularly. The purpose of this intensive cultivation is to loosen the soil structure to allow better drainage, faster root growth, aeration, incorporation of crop residues, and easier sowing of seed. Cultivation is also used to control weeds. Under typical practices—that is, when intensive tillage is combined with short rotations—fields are plowed or cultivated several times during the year, and in many cases this leaves the soil free of any cover for extended periods. It also means that heavy machinery makes regular and frequent passes over fields.

Ironically, intensive cultivation tends to degrade soil quality in a variety of ways. Soil organic matter is reduced as a result of accelerated decomposition and the lack of cover, and the soil is compacted by the recurring traffic of machinery. The loss of organic matter reduces soil fertility and degrades soil structure, increasing the likelihood of further compaction and making cultivation and its temporary improvements even more necessary. Intensive cultivation also greatly increases rates of soil erosion by water and wind.

In recent years, some farmers have turned to reduced-tillage or so-called no-tillage practices. No-till systems have reduced some of the negative impacts of intensive tillage, but as currently practiced they depend on herbicides for weed control. Since herbicide application has its own set of negative consequences (see Chemical Pest and Weed Control below) and because no-till systems reduce the input of organic material into the soil, this system is really just trading one set of problems for another.

Monoculture

Over the last century, agriculture all over the world has moved relentlessly toward specialization. Farming once meant growing a diversity of crops and raising livestock, but now farmers are far more likely to specialize, growing corn for livestock feed, for example, or raising hogs. In crop agriculture, specialization means monoculture—growing only one crop in a field, often on a very extensive scale. Monoculture allows more efficient use of farm machinery for cultivation, sowing, weed control, and harvest, and can create economies of scale with regard to purchase of seeds, fertilizer, and pesticides. Monoculture is a natural outgrowth of an industrial approach to agriculture, where labor inputs are minimized and technology-based inputs are maximized in order to increase productive efficiency. Monoculture techniques mesh well with the other practices of modern agriculture: monoculture tends to favor intensive cultivation, application of inorganic fertilizer, irrigation, chemical control of pests and weeds, and specialized plant varieties. The link with chemical pesticides is particularly strong; vast fields of the same plant are more susceptible to devastating attack by specific pests and diseases and require protection by pesticides. Many of the same problems occur when farmers plant large areas to organic monocultures.

Application of Synthetic Fertilizer

The spectacular increases in yields of the last half of the twentieth century were due in large part to the widespread and intensive use of synthetic chemical fertilizers. In the United States, the amount of fertilizer applied to fields each year increased rapidly after World War II, from 9 million tons in 1940 to more than 47 million tons in 1980. Although worldwide use of fertilizer increased most rapidly between 1950 and 1992, continuing increases in use since that period brought total world consumption of synthetic fertilizer beyond the 170-million-metric-ton mark in 2007 (FAOSTAT 2012).

Produced in large quantities at relatively low cost using fossil fuels, atmospheric nitrogen (N_2), and mined mineral deposits containing phosphorus (P), fertilizers can be applied easily and uniformly to crops to supply them with ample amounts of the most essential plant nutrients. Because they

meet plants' nutrient needs for the short term, fertilizers have allowed farmers to ignore long-term soil fertility and the processes by which it is maintained.

The mineral components of synthetic fertilizers, however, are easily leached out of the soil. In irrigated systems, the leaching problem may be particularly acute; a large amount of the fertilizer applied to fields actually ends up in streams, lakes, and rivers, where it causes **eutrophication** (excessive growth of oxygen-depleting plant and algal life). Fertilizer can also be leached into groundwater used for drinking, where it poses a significant health hazard. Use of nitrogen-based fertilizer is furthermore a problem for the atmosphere: it stimulates soil microbes to produce more nitrous oxide (N_2O), which acts as a greenhouse gas and depletes stratospheric ozone (Park et al. 2012). Finally, the cost of fertilizer is a variable over which farmers have no control since it rises with increases in the cost of petroleum.

Irrigation

An adequate supply of water is the limiting factor for food production in many parts of the world. Thus supplying water to fields from underground aquifers, reservoirs, and diverted rivers has been key to increasing overall yield and the amount of land that can be farmed. Although only 20% of the world's cropland is irrigated, this land produces 40% of the world's food (FAO 2011).

All sectors of society have placed rapidly increasing demands on freshwater supplies over the past half century, but agricultural purposes account for the lion's share of the demand—about 70% of water use worldwide (UN Water 2012). A clean, fresh, and sufficient supply of water has become a major issue on the immediate horizon not just for agriculture, but for all of human society (Pearce 2006). Unfortunately, agriculture is such a prodigious user of water that in many areas where land is irrigated for farming, irrigation has a significant effect on regional hydrology. The greatest problem is that groundwater is often pumped faster than it is renewed by rainfall. This overdraft can cause land subsidence, and near the coast it can lead to saltwater intrusion (Figure 1.1). In addition, overdrafting groundwater is essentially borrowing water from the future. Where water for irrigation is drawn from rivers, agriculture is often competing for water with water-dependent wildlife and urban areas. Dams built to hold water supplies have dramatic effects downstream on the ecology of rivers and block the spawning of anadromous fish. Irrigation has another type of impact as well: it increases the likelihood that fertilizers will be leached from fields and into local streams and rivers, and it can greatly increase the rate of soil erosion.



FIGURE 1.1 Furrow irrigation with gated pipe in coastal central California. Overdraft of the underground aquifers from which the irrigation water is pumped has caused saltwater intrusion, threatening the sustainability of agriculture in the region.

Chemical Pest and Weed Control

After World War II, chemical pesticides were widely touted as the new, scientific weapon in humankind's war against plant pests and pathogens. These chemical agents had the appeal of offering farmers a way to rid their fields once and for all of organisms that continually threatened their crops and literally ate up their profits. But this promise has proven to be false. Pesticides (i.e., insecticides, fungicides, and herbicides) can dramatically lower pest populations in the short term, but because they also kill pests' natural enemies, pest populations can often quickly rebound and reach even greater numbers than before. The farmer is then forced to use even more of the chemical agents. The dependence on pesticide use that results has been called the "pesticide treadmill." Augmenting the dependence problem is the phenomenon of increased resistance: pest populations continually exposed to pesticides are subjected to intense natural selection for pesticide resistance. When resistance among the pests increases, farmers are forced to apply larger amounts of pesticide or to use different pesticides, further contributing to the conditions that promote even greater resistance.

The metaphor of the "treadmill" is particularly apt because once a farmer gets on it, he or she finds it difficult to get off. With natural enemies eliminated from the system, ceasing to use pesticides is asking for serious crop damage. This is one reason why many farmers—especially those in developing nations—do not use other options, even though the problem of pesticide dependence is widely recognized. Even in the United States, the amount of pesticides applied to major field crops, fruits, and vegetables each year remains above 500,000 metric tons per year, more than twice the level it was in 1962, when Rachel Carson published *Silent Spring* (US EPA 2012). Pesticide resistance, the spread of insect pests and plant pathogens to regions where they had not previously existed, and the extensive use of genetically modified



FIGURE 1.2 Broadcast spraying to control codling moth in an apple orchard in the Pajaro Valley, CA.

(GM) crops *designed* to be grown in concert with intensive application of herbicides (see the next section) are all factors driving the worldwide increase in the use of chemical pest and disease controls. Ironically, total crop losses to pests have stayed fairly constant for the past 40–50 years despite increasing pesticide use (Pimentel 2005; Oerke 2006).

Besides costing farmers a great deal of money, pesticides can have a profound effect on the environment and on human health. Worldwide, millions of people every year experience symptoms of direct pesticide poisoning, and the ubiquitous presence of pesticides in water, soil, and food is implicated in increased incidence of cancer, reproductive and developmental disorders, and other maladies. Pesticides applied to fields kill beneficial insects and those essential to natural system food webs, and they are easily washed and leached into surface water and groundwater, where they enter the food chain, affecting animal populations at every level and often persisting for decades.

Manipulation of Plant and Animal Genomes

Humans have selected for specific characteristics among crop plants and domesticated animals for thousands of years; indeed, human management of wild species was one of the foundations of the beginning of agriculture. In recent decades, however, technological advances have brought about a revolution in the manipulation of genes. First, advances in breeding techniques allowed for the production of hybrid seeds, which combine the characters of two or more plant strains. Hybrid plant varieties can be much more productive than similar nonhybrid varieties and have thus been one of the primary factors behind the yield increases achieved during the so-called "green revolution." The hybrid varieties, however, often require optimal conditions—including intensive application of inorganic fertilizer—in order to realize their productive potential, and many require pesticide application to protect them from extensive pest damage because they lack the pest resistance of their nonhybrid cousins. In addition, hybrid plants cannot produce seeds with the same genome as their parents, making farmers dependent on commercial seed producers.

More recently, geneticists have developed techniques that allow them to splice genes from a variety of organisms into target genomes to create "customized" plant and animal varieties. These organisms are referred to as **transgenic**, GM, or **genetically engineered** (GE).

Only a few animal species used for food have been genetically engineered as yet—these include pigs with spinach genes that produce lower-fat bacon, cows that produce milk with higher casein levels, and salmon that grow at twice the rate of their wild kin—but transgenic crop plants have become widespread and very important in agricultural production. Between 1996 and 2012, the area planted to GE crops worldwide increased 100-fold, from 1.7 million hectares to over 170 million hectares, making "biotech" crops "the fastest adopted crop technology in the history of modern agriculture" (James 2012). Although developed countries have long been the leaders in production of biotech crops—69.5 million hectares were planted in the United States in 2012, for example—developing countries are now adopting the crops at a faster rate. The area planted to biotech crops in developing countries surpassed that in developed countries in 2012.

Two types of GM crops have become particularly prevalent: those engineered to be tolerant of herbicides and those containing genes directing the plants to produce the same insecticidal toxins produced by the bacterium *Bacillus thuringiensis* ("Bt crops"). Herbicide-tolerant crops are designed to be treated with herbicides—usually glyphosate—to kill weeds but not the crop plants; Bt crops protect themselves from herbivory, reducing the need for insecticides. Together, these crops account for about 95% of the acres planted to cotton and soybean in the United States and about 85% of the acres planted to corn (Benbrook 2012).

Although GE organisms hold many promises—reducing the use of pesticides and irrigation, allowing agriculture on soils too saline for normal crops, and increasing the nutritional value of some crops—there are many concerns about the spread of this and related biotechnologies. One main source of concern is the potential for the migration of modified genes into other populations, both wild and domestic. This could result, for example, in more aggressive weeds or the introduction of toxins into crop plants. In the case of the modified salmon mentioned earlier, the fish could easily escape and cross with wild salmon, possibly upsetting ocean food chains. Increased use of transgenic crops may also diminish agrobiodiversity, as traditional cultivars are abandoned, and increase the dependence of farmers on the transnational corporations owning the patents on the new organisms.

The rapid rise to dominance of herbicide-tolerant and Bt crops in US agriculture has revealed what may be one of the most serious drawbacks of GM food organisms: in both cases, the target pests—weeds for herbicide-tolerant crops and insects for Bt crops—become resistant, creating yet another version of the pesticide treadmill. Many weed

species have quickly become resistant to glyphosate, forcing farmers planting herbicide-resistant crops to increase application of glyphosate, spray it more often, and add in other herbicides that have a different mode of action (and are often more toxic to humans). Because of this "super weed" phenomenon, herbicide use in the United States increased exponentially between 1996 and 2011; US farmers now use 527 million more pounds of herbicide than they did in 1996 (Benbrook 2012). Although Bt crops have, during the same period, reduced insecticide use moderately, insect pests have become increasingly resistant to Bt toxins, causing farmers to bring back the insecticides they used in the past in order to preserve the efficacy of Bt technology. Moreover, the large amounts of Bt toxin produced by Bt crops cause the toxin to appear in ever-higher amounts in animal feed, human food, and the environment.

Factory Farming of Animals

If you live in a developed country, a large portion of the meat, eggs, and milk that you eat probably comes from large-scale, industrialized operations driven by the goal of bringing these food products to market at the lowest possible unit cost. The animals in these "confined animal feeding operations" (CAFOs) are typically crowded so tightly they can barely move, given antibiotics to prevent the spread of disease, and fed highly processed soy- and corn-based feed supplemented with hormones and vitamins. Even though they are completely dependent on crop agriculture for the production of feed, CAFOs are disconnected—spatially and functionally—from the fields in which the feed grains are grown (Figure 1.3).

Factory-farm livestock production is another manifestation of the specialization trend in agriculture. In many ways, factory farming is for pigs, cattle, and poultry while monoculture is for corn, wheat, and tomatoes. The livestock in CAFOs are more susceptible to disease, just as monocropped corn plants are to pest damage, and both require chemical inputs (pharmaceuticals for livestock and pesticides for crops) to compensate. Both factory farming and monoculture encourage the use of organisms bred or engineered for productive efficiency and dependent on the artificial conditions of the industrial process.

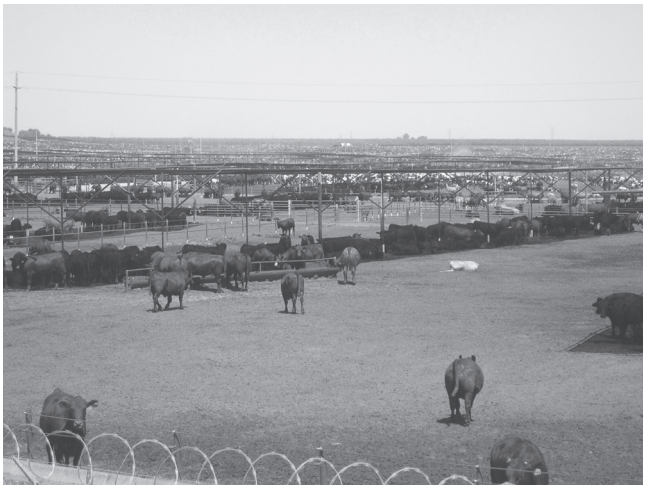


FIGURE 1.3 A CAFO in California's Central Valley.

Factory farming is criticized by animal rights groups as cruel and inhumane. Laying hens and broiler chickens are routinely debeaked to keep them from pecking each other; hogs are often kept in pens so small they cannot turn around; beef cattle commonly suffer slow and painful deaths at the slaughterhouse.

There are many other reasons to be critical of the industrial approach to raising livestock. CAFOs, for example, have serious impacts on the environment. Disposal of the massive amounts of manure and urine generated by the confined animals is a huge problem, usually dealt with by treating the wastes in large anaerobic lagoons that leak nitrates into surface streams and groundwater and allow ammonia to escape into the atmosphere. This problem arises because CAFOs by their very nature cannot recycle nitrogen within the system, as is the case on smaller traditional farms where animals and crop plants are raised together. Thus nitrogen becomes a problematic waste product instead of a valuable plant nutrient.

The rise in factory farming is coupled with a worldwide trend toward diets higher in meat and animal products. As demand for meat increases, industrialized methods of animal food production become more profitable and more widespread, replacing more sustainable pastoral and mixed crop–livestock systems.

WHY INDUSTRIAL AGRICULTURE IS NOT SUSTAINABLE

The practices of industrial agriculture all tend to compromise future productivity in favor of high productivity in the present. The ways in which industrial agriculture puts future productivity at risk are many. Agricultural resources such as soil, water, and genetic diversity are overdrawn and degraded, global ecological processes on which agriculture ultimately depends are altered, human health suffers, and the social conditions conducive to resource conservation are weakened and dismantled. In economic terms, these adverse impacts are called **externalized costs**. Because their consequences can be temporarily ignored or absorbed by society in general, they are excluded from the cost–benefit calculus that allows industrial agricultural operations to continue to make economic sense.

An important feature of industrial agriculture's externalized costs is that they have serious consequences both for the future and the present. These "unsustainable" aspects of industrial agriculture are not problematic just because they are unsustainable—because they will one day cause the system to collapse—but because they are causing, in the present, real human suffering and irreparable damage to the ecological systems on which we rely. They are also problematic because when they do begin to pull industrial agriculture into a state of crisis, agriculture would not be the only part of human society that will be impacted.

Soil Degradation

Every year, according to the Food and Agriculture Organization (FAO) of the United Nations (UN), between five and seven million hectares of valuable agricultural land are lost to soil degradation. Other estimates run as high as 10 million hectares per year (e.g., World Congress on Conservation Agriculture 2005). In 2011, the FAO estimated that 33% of the earth's land is highly or moderately degraded, with the majority of this land in areas with high poverty rates (FAO 2011). Degradation of soil can involve salting, waterlogging, compaction, contamination by pesticides, decline in the quality of soil structure, loss of fertility, and erosion by wind and water.

Although all these forms of soil degradation are severe problems, erosion is the most widespread. Under optimal conditions, soil is created at the rate of about 1 ton/ha/year, but worldwide soil is being eroded from industrially farmed land at a rate one to two orders of magnitude greater (Montgomery 2007). This means that in just a short period, humans have wasted soil resources that took thousands of years to be built up (Figure 1.4).

The cause–effect relationship between industrial agriculture and soil erosion is direct and unambiguous. Intensive tillage, combined with monoculture and short rotations, leaves the soil exposed to the erosive effects of wind and rain. The soil lost through this process is rich in organic matter, the most valuable soil component. Similarly, irrigation is a direct cause of much water erosion of agricultural soil.

Combined, soil erosion and the other forms of soil degradation render much of the agricultural soil of the world increasingly less fertile. Some land—severely eroded or too salty from evaporated irrigation water—is lost from production altogether. The land that can still produce is kept productive by the artificial means of adding synthetic fertilizers. Although fertilizers can temporarily replace lost nutrients, they cannot rebuild soil fertility and restore soil health; moreover, their use has a number of negative consequences, as discussed earlier.

Since the supply of agricultural soil is finite, and because natural processes cannot come close to renewing or restoring soil as fast as it is degraded, agriculture cannot be sustainable until it can reverse the process of soil degradation. Current agricultural practices must undergo a vast change if the precious soil resources we have remaining are to be conserved for the future.



FIGURE 1.4 Severe soil erosion on a sloping hillside following intense winter rains. In this strawberry-growing region in the Elkhorn Slough watershed of central California, soil losses exceed 150 tons/acre in some years.

Overuse of Water and Damage to Hydrological Systems

Freshwater is becoming increasingly scarce in many parts of the world as industry, expanding cities, and agriculture compete for limited supplies. Some countries have too little water for any additional agricultural or industrial development to occur. To meet demands for water in many other places, water is being drawn from underground aquifers much faster than it can be replenished by rainfall, and rivers are being drained of their water to the detriment of aquatic and riparian ecosystems and their dependent wildlife. Many of the world's major rivers—including the Colorado, Ganges, and Yellow—now run dry for part of the year as a result.

Agriculture accounts for more than 70% of global water use. Most of this water is used to irrigate crops. For the most part, irrigation is employed not to make land productive, but to make it *more* productive. The 20% of agricultural land worldwide that is irrigated produces about 40% of the world's food supply (FAO 2011). To generate this considerable increase in yield beyond what would otherwise be the case, irrigated agriculture uses tremendous volumes of water.

Irrigated agriculture uses so much water in part because it uses water wastefully. More than half of the water applied to crops is never taken up by the plants it is intended for (Van Tuijl 1993). Instead, this water either evaporates from the soil surface or drains out of fields. Some wastage of water is inevitable, but a great deal of waste could be eliminated if agricultural practices were oriented toward conservation of water rather than maximization of production. For example, crop plants could be watered with drip irrigation systems, and production of water-intensive crops such as rice could be shifted away from regions with limited water supplies.

The increasing importance of meat in human diets worldwide is another factor in agriculture's rising demand for water, as is the trend toward concentrated grain feeding of livestock. Animal factories use a great deal of water for cooling the animals and flushing their wastes, and many animals drink large amounts of water. Hogs, for example, can consume up to 8 gal/animal/day (Marks and Knuffke 1998). And these are

just the direct uses of water for raising livestock. Factoring in the water needed to grow the biomass fed to animals, animal-derived food requires at least twice as much water to produce as plant-derived food, and usually much more. The difference between the amount of water needed to grow calorie-equivalent amounts of plant food and animal food can be extreme. For example, it takes only 89 L of water to grow 500 cal of potatoes, but an astonishing *fifty-five times* more, or 4902 L, to raise 500 cal of grain-fed beef (Postel and Vickers 2004). If we look at protein alone, the ratio is even more skewed: on average, producing 1 kg of animal protein requires about 100 times as much water as producing 1 kg of grain protein (Pimentel and Pimentel 2003).

In addition to using a large share of the world's freshwater, industrial agriculture has an impact on regional and global hydrological patterns and the aquatic, riparian, and marine ecosystems dependent on them. First, by drawing such large quantities of water from natural reservoirs on land, agriculture has caused a massive transfer of water from the continents to the oceans. A 2012 study concluded that an observed sea level rise of 0.77 mm/year between 1961 and 2003, about 42% of the total rise, was due to the transfer of water from on-land storage basins to the sea. Most of this transfer is due to the use of underground aquifers for irrigation (Pokhrel et al. 2012). Moreover, the amount of water that agriculture causes to be moved from the land to the oceans is only increasing as more land is brought under irrigation. Second, where irrigation is practiced on a large scale, agriculture brings about changes in hydrology and microclimate. Water is transferred from natural watercourses to fields and the soil below them, and increased evaporation changes humidity levels and may affect rainfall patterns. These changes in turn significantly impact natural ecosystems and wildlife. Third, the dams, aqueducts, and other infrastructure created to make irrigation possible have dramatically altered many of the world's rivers, causing enormous ecological damage. Rivers that once provided valuable ecosystem services to human society cannot do so anymore—their wetland, aquatic, and floodplain ecosystems can no longer absorb and filter out pollutants or provide habitat for fish and waterfowl, and they can no longer deposit the rich sediment so important for restoring the fertility of agricultural soils in floodplain areas (Figure 1.5).

Agriculture's large and growing use of water will only grow more serious as a fundamental issue facing humankind. As the demand for water increases, the guarantee of an adequate supply becomes less and less assured because climate change is reducing mountain snowfall, melting high-altitude glaciers, increasing the frequency of droughts, causing salinization of groundwater in coastal areas, and degrading the ecosystem processes that help purify water. If industrial agriculture continues to use water in the same ways, our rivers will become increasingly crippled and regional water crises will become increasingly common, either shortchanging the environment, marginalized peoples, and future generations or limiting irrigation-dependent food production.



FIGURE 1.5 The San Luis Dam in California. Built in part to hold irrigation water for farms on the west side of the San Joaquin Valley, it is one of an estimated 800,000 dams in the world that trap life-giving silt, destroy riverine and riparian ecosystems, and completely alter natural hydrological functioning.

Pollution of the Environment

More water pollution comes from agriculture than from any other single source. Agricultural pollutants include pesticides, herbicides, other agrochemicals, fertilizer, animal wastes, and salts.

Pesticides and herbicides—applied in large quantities on a regular basis, often from aircraft—are easily spread beyond their targets, killing beneficial insects and wildlife directly and poisoning farmers and farmworkers. The pesticides that make their way into streams, rivers, and lakes—and eventually the ocean—can have serious deleterious effects on aquatic ecosystems. They can also affect other ecosystems indirectly. Fish-eating raptors, for example, may eat pesticide-laden fish, reducing their reproductive capacity and thereby impacting terrestrial ecosystems. Although persistent organochloride pesticides such as DDT—known for their ability to remain in ecosystems for many decades—are being used less in many parts of the world, their less-persistent replacements are often much more acutely toxic.

Pesticides also pose a significant human health hazard. They spread throughout the environment by hydrological, meteorological, and biological means, and so it is impossible for humans to avoid exposure. In its 2003 edition of *Human Exposure to Environmental Chemicals*, the Centers for Disease Control (CDC) reported that all of the 9282 people they tested had pesticides and their breakdown products in their bodies, and the average person had detectable amounts of 13 different pesticides (Schafer et al. 2004). Similar incidences of exposure and detection were reported in the CDC's 2013 report (CDC 2013). Pesticides enter our bodies through our food and our drinking water. In one study (Gilliom and Hamilton 2006), pesticide contamination was detected in 97% of streams tested in agricultural and urban areas, in 94% of streams tested in areas with mixed land use, and in 65% of

streams tested in undeveloped areas. Pesticides were found in 61% of groundwater samples in agricultural areas and 55% of samples in urban areas. Another study (Wu et al. 2010) found that the herbicide atrazine, which is used very commonly for corn production, was present in 75% of all watersheds and 40% of the drinking water wells in corn-producing regions of the United States, and estimates that over 33 million people in the United States have been exposed to atrazine in their drinking water. If all the drinking water sources in the United States at risk for pesticide contamination were properly monitored for the presence of harmful agents, the cost would be well over $15 billion (Pimentel 2005).

Fertilizer leached from fields is less directly toxic than pesticides, but its effects can be equally damaging ecologically. In aquatic and marine ecosystems it promotes the overgrowth of algae, causing eutrophication and the death of many types of organisms. Nitrates from fertilizers and livestock manure are also a major contaminant of drinking water in many areas. When nitrates enter aquifers they are not easily removed, and frequently alternative drinking water sources are not available. As a result, many people in agricultural regions are exposed to nitrate levels in excess of established safe thresholds and have an increased risk of cancer and reproductive disorders. Rounding out the list of pollutants from croplands are salts and sediments, which in many locales have degraded streams, helped destroy fisheries, and rendered wetlands unfit for bird life.

Where factory farming has become the dominant form of meat, milk, and egg production, animal waste has become a huge pollution problem. Farm animals in the United States produce far more waste than do humans. The large size of feedlot and other factory farming operations poses challenges for the treatment of these wastes. As noted in the previous text, the wastes are typically treated in large anaerobic lagoons not well suited to protection of the environment. Some of the nitrogen from the wastes leaks out of the lagoons and into underlying aquifers, adding large quantities of nitrates to the groundwater and eventually to rivers. Even more nitrogen from the wastes converts to ammonia and enters the atmosphere, where it combines with water droplets to form ammonium ions. As a result, the rainwater downwind of livestock feeding operations often has extremely high concentrations of ammonium ions. Although most treated animal waste is ultimately applied to fields as fertilizer, the phosphorus and nitrogen it contains are beyond useful levels for most crops. Furthermore, factory farms often have so much waste to get rid of that they apply more treated waste to fields than the soil can accommodate, and do so year-round, even at times in the crop cycle when fields and crops are unable to absorb it. The excess nitrogen and phosphorus find their way into streams, rivers, and the local drinking water supply.

Through all these various avenues, tons of nitrogen and phosphorus from animal waste and inorganic fertilizer make their way into waterways and then into the oceans, creating large "dead zones" near river mouths. More than 50 of these dead zones exist seasonally around the world, with some of the largest—in the Chesapeake Bay, Puget Sound, and Gulf of Mexico—off the coast of the United States. In the summer of 2013, the dead zone in the Gulf of Mexico reached a record-breaking size of more than 8000 square miles.

Destruction of Natural Habitat

Farming entails the conversion of native vegetation—the habitat for native species of insects, birds, mammals, and other animals—into land intensively managed by humans. That is the nature of agriculture and the price of supporting large populations of human beings on the earth. But different forms of agriculture have vastly different impacts on native vegetation and natural habitat. As will be discussed in Chapter 23, land managed by humans for food production can support healthy populations of beneficial insects, birds, and other vertebrates and invertebrates, serving in this regard as a reasonable substitute for the natural habitats that once existed on the land. For a variety of reasons, industrial agriculture has proven remarkably effective at not only eliminating vast expanses of native vegetation but also at essentially sterilizing agricultural land and reducing its habitat value to essentially zero.

Industrial agriculture supports a drive to convert as much natural habitat as possible to farmland because more land in production means more profit. More often than not, farmers expand their areas of production not to grow more food for people, but to grow more corn and other agricultural commodities for biofuel production and animal feed. In the United States, conversion of additional land to corn production has been directly linked to a rise in the price of corn, which is a product of federal subsidies for biofuel production.

All the practices of industrial agriculture described earlier combine to make the large bulk of cropland in many areas essentially worthless as wildlife habitat. Intensively tilled monocultures of genetically uniform crops fertilized with inorganic fertilizers can serve as a habitat for very few animals except insect pests, and in attempting to control these pests with pesticides, industrially oriented farmers insure that other insects are eliminated as well. More recently, the development of herbicide-resistant crop varieties has allowed farmers to escalate their war against weeds to a new level, creating vast stretches of agricultural landscape with no refuges for beneficial insects and no food plants for migrating populations of butterflies.

The effects of eliminating natural vegetation and reducing the habitat value of agricultural land may be slow to accumulate, but there is little doubt that they may become severe. Some of the effects will be felt directly by agroecosystems, as pollinators such as European and native bees become scarce and reductions in populations of natural enemies of insect pests make farmers more dependent on pesticides. But even more worrisome are the larger-scale effects, which include precipitous declines in biological diversity and deterioration of ecosystems that provide farmers and other humans with critical ecosystem services (such as water purification, buffering of floods, groundwater recharge, and erosion control).



FIGURE 1.6 Pine forest habitat on the Florida Piedmont being bulldozed to make way for irrigated corn and pasture for cattle production. The white sandy soil exposed by forest clearing is low in organic matter and nutrients and requires significant external inputs to support agricultural production. The agroecosystems put in place here will have a fraction of the diversity of the undisturbed forest.



FIGURE 1.7 Equipment yard of a large industrial vegetable farm in the Salinas Valley, CA. High levels of external inputs are needed to level, rip, and cultivate soil, plant seeds or transplant seedlings, apply fertilizers, spray pesticides, irrigate, and harvest crops such as the monoculture broccoli seen in the foreground.

Dependence on External Inputs and Nonrenewable Resources

Industrial agriculture has achieved its high yields mainly by increasing agricultural inputs. These inputs comprise physical factors of production such as irrigation water, fertilizer, pesticides, and processed feed and antibiotics; the energy used to manufacture these substances, to run farm machinery and irrigation pumps, and to climate-control animal factories; technology in the form of hybrid and transgenic seeds, new farm machinery, and new agrochemicals; and knowledge in the form of the expertise needed to use and manage these inputs. These inputs all come from outside the agroecosystem itself; their extensive use has consequences for farmers' profits, use of nonrenewable resources, and the locus of control of agricultural production (Figure 1.7).

The longer industrial practices are used on farmland, the more the system becomes dependent on external inputs. As intensive tillage and monoculture degrade the soil, continued fertility depends more and more on the input of fossil-fuel-derived nitrogen fertilizer and other nutrients. And using reduced-tillage systems to limit the problems caused by intensive tillage does nothing to break this dependency because it usually requires that intensive herbicide use take the place of tillage as a weed control method.

Agriculture cannot be sustained as long as this dependence on external inputs remains. First, the natural resources from which many of the inputs are derived are nonrenewable and their supplies finite. Second, dependence on external inputs leaves farmers, regions, and whole countries vulnerable to supply shortages, market fluctuations, and price increases. In addition, excessive use of inputs has multiple negative off-farm and downstream impacts, as noted earlier.

The most notable of external inputs in industrial agriculture is fossil fuels. The dependence of industrial agriculture on fossil fuels has become so extreme—they are critical for everything from manufacture of nitrogen fertilizer to transport of food from one side of the globe to the other—that food prices have become correlated directly with energy prices. Although agriculture's dependence on an input that will eventually be used up is a cause for concern, a continued flow of fossil fuels has been guaranteed for the medium term by the development of new extractive technologies such as "fracking" and the exploitation of deeper offshore oil fields. The same thing cannot be said, however, for another critical external input: phosphorus. Mined deposits of phosphorus-rich minerals—the sole source of this important macronutrient in synthetic fertilizer—may be mostly used up within the next four decades.

Production of Greenhouse Gases and Loss of Carbon Sinks

As an economic sector, agriculture is the third largest contributor to greenhouse gas emissions worldwide, behind transportation and the burning of fossil fuels for power and heat. Although it is impossible to grow, process, and distribute food without releasing carbon dioxide and other greenhouse gases into the atmosphere, our present food system makes a much larger contribution to climate change than it would if organized according to agroecological principles. The geographic and economic separation between farmers and consumers insures the burning of large quantities of fossil fuels to distribute and transport food; input-intensive monoculture requires that fossil fuels be used to produce and distribute inorganic fertilizers, pesticides, and other inputs and that farmers be dependent on fossil-fuel-consuming field equipment. Further, industrial agriculture's primary focus on the maximization of yield and profit gives farmers little motivation to use fossil-fuel energy and the inputs derived from it efficiently. It is common, for example, for farmers to

apply excess nitrogen fertilizer, much of which ends up as the greenhouse gas nitrous oxide.

The food system's focus on production of meat and dairy products is a major reason why agriculture produces so much greenhouse gas. Approximately 37% of agriculture's total greenhouse gas emissions—in the form of the potent greenhouse gas methane—come from the digestive systems of the world's livestock. Livestock are also responsible for much of agriculture's emission of carbon dioxide and nitrous oxide. The nitrous oxide comes from bacterial processing of the nitrogen in livestock manure; the carbon dioxide comes from the rapid decomposition of crop residue in the tilled fields used to produce livestock feed.

In addition to producing greenhouse gases, industrial agriculture exacerbates climate change by reducing the ability of the biosphere to hold carbon in a fixed, organic form. At any particular moment, a significant portion of the carbon in circulation—that is, not locked away in geologic structures below the surface—is not in gaseous form in the atmosphere, but present as dissolved CO_2 in the oceans and in organic or mineral form in earth's terrestrial ecosystems. This latter "sink" of carbon is largely made up of vegetative biomass and the microbial biomass, humus, and organic and mineral carbon of the soil. Industrial agriculture involves practices (described in Practices of Industrial Agriculture section) that reduce the storage capacity of both of these terrestrial carbon sinks. Much of this occurs in the clearing of large tracts of woody vegetation—much of it tropical rainforest—for pasture land and for growing livestock feed, palm oil, and biofuel feedstock. Additionally, intensive tillage, application of inorganic fertilizer, and a strong reliance on annual crops dramatically reduce the ability of agricultural soils to sequester and store carbon because they reduce the soil's biological activity and expose its organic matter to depletion by erosion, chemical degradation, and bacterial respiration.

In these many ways, industrial agriculture makes a significant contribution to climate change, thereby playing a role in making much of the earth less hospitable to agriculture in any form.

Loss of Genetic Diversity

Throughout most of the history of agriculture, humans have increased the genetic diversity of crop plants and livestock worldwide. We have been able to do this both by selecting for a variety of specific and often locally adapted traits through selective breeding, and by continually recruiting wild species and their genes into the pool of domesticated organisms. In the last 100 years or so, however, the overall genetic diversity of domesticated plants and animals has declined. Many varieties of plants and breeds of animals have become extinct, and a great many others are heading in that direction. About 75% of the genetic diversity that existed in crop plants in 1900 had been lost 100 years later (Nierenberg and Halweil 2004). The UN FAO reported in 2010 that even though modern breeding programs are continually releasing new varieties for use in production, the observed trend is for farmers (and especially traditional farmers in developing countries) to abandon their locally adapted varieties (FAO 2010). The UN FAO estimated in 1998 that as many as two domesticated animal breeds were being lost each week worldwide (FAO 1998), and noted again in 2007 that a similar rate of loss continued as more farmers shifted to market-oriented confinement production systems, putting at least 20% of known animal breeds in danger of extinction (FAO 2007).

In the meantime, the genetic bases of most major crops and livestock species have become increasingly uniform. At the end of the last century, only six varieties of corn, for example, accounted for more than 70% of the world's corn crop, and 99% of the turkeys raised in the United States belonged to a single breed (FAO 1998).

The loss of genetic diversity has occurred mainly because of industrial agriculture's emphasis on short-term productivity gains. When highly productive varieties and breeds are developed, they tend to be adopted in favor of others, even when the varieties they displace have many desirable and potentially desirable traits. Genetic homogeneity among crops and livestock is also consistent with the maximization of productive efficiency because it allows standardization of management practices.

For crop plants, a major problem with increasing genetic uniformity is that it leaves each crop as a whole more vulnerable to attack by pests and pathogens that acquire resistance to pesticides and to the plants' own defensive compounds; it also makes crops more vulnerable to changes in climate and other environmental factors. These are not insignificant or hypothetical threats. Every year, crop pests and pathogens destroy an estimated 30%–40% of potential yield. Plant pathogens can evolve rapidly to overcome crop's defenses, and global commerce and genetically uniform farm fields allow these new virulent strains to spread rapidly from field to field and continent to continent. In a report on crop diversity and disease threats released in 2005, researchers identified four diseases with the potential to devastate the US corn crop, five that could threaten potatoes, and three with the potential to harm US-grown wheat (Qualset and Shands 2005). In late 2004, a new soybean rust (a type of fungus) appeared in the southern United States and began to attack the soybean crop. By 2009, soybean rust had spread to 16 states and more than 576 counties in the United States and at least 3 states in Mexico. By 2012, the US Department of Agriculture (USDA) had reported its appearance in Texas and Florida. None of the commercial soybean varieties planted in the United States are yet resistant to the rust fungus, and scientists are concerned about the potential impact on the multibillion US dollar soybean harvest as the rust continues to spread.

Throughout the history of agriculture, farmers—and more recently, plant scientists—have responded to outbreaks of disease by finding and planting resistant varieties of the affected crop. But as the size of each crop's genetic reservoir declines, there are fewer and fewer varieties from which to

draw resistant or adaptive genes. The importance of having a large genetic reservoir can be illustrated by example. In 1968, greenbugs attacked the US sorghum crop, causing an estimated $100 million in damage. The next year, insecticides were used to control the greenbugs at a cost of about $50 million. Soon thereafter, however, researchers discovered a sorghum variety that carried resistance to the greenbugs. No one had known of the greenbug resistance, but it was there nonetheless. This variety was used to create a hybrid that was grown extensively and not eaten by greenbugs, making the use of pesticides unnecessary. Such pest resistance is common in domesticated plants, "hiding" in the genome but waiting to be used by plant breeders. As varieties are lost, however, the valuable genetic reservoir of traits is reduced in size, and certain traits potentially invaluable for future breeding are lost forever. There may very well be a soybean variety somewhere in the world resistant to the new soybean rust, but will plant scientists locate it before it goes extinct? A broader issue is that agricultural systems with narrowed genetic bases are less effective in integrating with and supporting the function of natural systems and thereby helping to create multifunctional landscapes (see Chapter 23).

Increasing vulnerability to disease is also a serious concern for domesticated animal species as they lose their genetic diversity, but perhaps more serious is increased dependence on methods of industrial food production. Livestock breeds that are not adapted to local conditions require climate-controlled environments, doses of antibiotics, and large amounts of high-protein feed.

Loss of Local Control over Agricultural Production

Accompanying the concentration of agriculture into large-scale monocultural systems and factory farms has been a dramatic decline in the number of farms and farmers, especially in developed countries where mechanization and high levels of external inputs are the norm. From 1920 to the turn of the century, the number of farms in the United States dropped from more than 6.5 million to just over 2 million, and the percentage of the population that lived and worked on farms dropped below 2%. Data from the 2000 US census showed that only 0.4% of the employed civilians in the United States listed their occupation as "farmer or rancher" (US Census Bureau 2005). Although the 2007 agricultural census showed the first increase in the number of farms in more than 30 years, the increase was primarily in large- and small-scale operations. Midsize farms have continued to decline in the United States at much the same rate as seen during the last century (Gliessman 2009).

In developing countries as well, rural people who work primarily in agriculture continue to abandon the land to move to urban and industrial areas, which will hold an estimated 60% of the world's population by 2030, and perhaps 70% by 2050. China is now carrying out a long-term plan to move 250 *million* rural people—most of them small-scale farmers—into newly built towns and cities. The country's leaders hope that expanding the number of urban dwellers will greatly increase consumption and thus economic growth, but they have not directly addressed the issue of how those in the cities will be fed, or what the effects will be of leaving responsibility for food production in the hands of fewer and fewer people. As shown in Figure 1.8, there are now far more people in the world whose livelihoods are non-agricultural than there are people who grow food, and this gap continues to widen over time.

Besides encouraging an exodus from rural areas, large-scale commodity-oriented farming tends to wrest control of food production from rural communities. This trend is disturbing because local control and place-based knowledge and connection are crucial to the kind of management required

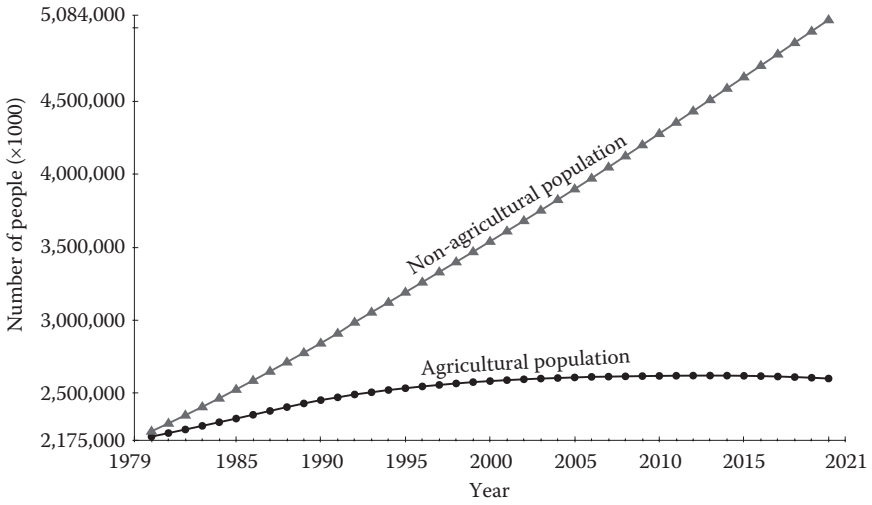


FIGURE 1.8 Number of people worldwide involved in agriculture and not involved in agriculture. (Data from FAOSTAT, Food and Agriculture Organization of the United Nations, Statistics database, 2013, http://faostat3.fao.org/home/index.html, Dates of access range from January 1, 2013 to December 31, 2013). Figures for 2011 and beyond are projections.

for sustainable production. Food production carried out according to the dictates of the global market, and through technologies developed elsewhere, inevitably severs the connection to ecological principles. Experience-based management skill is replaced by purchased inputs requiring more capital, energy, and use of nonrenewable resources. Farmers become mere instruments of technology application, rather than independent decision makers and managers.

Smaller-scale farmers seem to have little power against the advancement of industrial agriculture. Smaller farms cannot afford the cost of upgrading their farm equipment and technologies in order to compete successfully with the large farm operations. Moreover, the increase in the share of the food dollar going to distributors and marketers, coupled with cheap food policies that have kept farm prices relatively stable, has left many farmers in a tightening squeeze between production costs and marketing costs. As the industrial food system has expanded in the United States over the last century, increasing the physical and economic distance between farmers and consumers, US farmers' share of the consumer food dollar has continued to shrink, and now stands below $0.16 according to the USDA (Economic Research Service 2014).

Faced with such economic uncertainty, there is less incentive for farmers to stay on the land. One trend is for larger farmers to buy out their smaller neighbors. But when agricultural land is adjacent to rapidly expanding urban centers, such as in California, the incentive instead is to sell farmland at the inflated value it has as urban land. Because of this dynamic, the agriculturally rich Great Central Valley of California has seen the loss of hundreds of thousands of hectares of farmland to development since 1950, and the rate of loss of agricultural land in the state as a whole averaged 49,700 acres annually from 1988 to 1998 (Kuminoff et al. 2001). Of the 538,000 acres of agricultural land in California urbanized since the Gold Rush, one-sixth has been lost to farming in the few decades since 1990 (Thompson 2009). At present, agricultural land is lost to urbanization at the rate of more than 40,000 acres annually (American Farmland Trust 2007).

In less-developed countries, the growth of large-scale export agriculture has an even more ominous effect. Elites in these countries have, for a long time, gained control of land through various and often illegal means to increase production of export crops. More recently, however, the growing value of agricultural land in less-developed countries has attracted international investors, who have been buying it up at a rapid pace. In the decade between 2000 and 2010, more than 203 million hectares of land in less-developed countries were the object of sale or lease negotiations (Anseeuw et al. 2012). The majority of these land deals were made for the express purpose of growing export crops—biofuels in particular—and will contribute nothing to the food supplies of the countries in which they are located. In nearly all cases, realizing investors' plans means removing the people living on and farming the land, often violently and usually without consultation or compensation (Geary 2012).

As a result of these and other trends, rural people—once able to feed themselves adequately *and* sell surplus food to city dwellers—now make up the most food-insecure group worldwide. It is estimated that 80% of the world's hungry live in rural areas (Mikhail 2012). And as more and more rural smallholders are pushed off the land, they migrate to cities, where they become dependent on others for their food. Since more of the food produced in the countryside is destined for export, increasing amounts of food for the expanding urban areas must be imported. In 2009, 111 developing countries in the world, most of them with low average incomes, were classified as "net food importers" (Valdéz and Foster 2012). This imbalance threatens the food security of less-developed countries and makes their people extremely vulnerable to spikes in the prices of globally traded food staples.

Increasing Vulnerability and Risk

The size, scale, integration, and technological sophistication of the world food system tends to give the impression that it can easily resist the environmental vagaries—droughts, floods, cold snaps, pest infestations—that have plagued farmers since humans took up agriculture thousands of years ago. But this impression is a false one: industrial agriculture has actually made itself extraordinarily vulnerable to extreme weather events, climatic shifts, and pests and diseases (Figure 1.9).

A central cause of this vulnerability is the practice of monoculture, especially when it is combined with its usual concomitant of increasing the genetic uniformity of the crop. Planting the same variety of a single crop across a wide geographic area virtually assures that when nature serves up conditions hostile to that crop's development—a late spring frost, a severe drought, an extreme weather event—the damage will be widespread. When the damage is caused by drought, the effects are intensified by the dependence on



FIGURE 1.9 A dried-up, drought-affected soybean field in northern Iowa. The extreme drought of 2012, one of the most severe on record, caused significant crop losses in the Midwest of the United States. (Photo courtesy of Laura Jackson.)

synthetic fertilizer, because years of providing crop nutrition solely through chemical means have dramatically lowered the soil's moisture-holding capacity through depletion of its organic matter. As noted earlier, monoculture and genetic uniformity also dramatically increase vulnerability to pests and disease. A virtual sea of host organisms, all with their natural resistance bred out of them, is the perfect opportunity for a fungus, virus, or insect to vastly improve its reproductive success in a very short time span. Further exacerbating the problem is the inherent risk of depending on only three crops—corn, rice, and wheat—for more than half of the world's food.

Climate change assures that industrial agriculture's vulnerability (or, put the other way, its lack of resilience) will increasingly become a matter of serious concern. Climate change is likely to increase the frequency and severity of droughts and floods, to increase the incidence of extreme cold and heat, to reduce the mountain snowfall on which many regions rely for irrigation water, and to allow pests and diseases to move to regions where they were formerly excluded by winter cold. An earth beset by a changing climate needs exceptionally resilient agroecosystems, not the opposite.

Because of its interconnected nature, the world food system is also vulnerable to social, political, and economic factors that have no direct connection to climate, weather, or the environment. Increases in the price of oil, trade agreements, unilateral governmental actions, and disruptions in the world economy are among the many factors that may have important effects on food prices and food supplies. In this realm, however, it is necessary to clarify who bears the brunt of the "vulnerability." Industrial agriculture has become so deeply integrated into the world economic system, which is controlled by a relative handful of elites, that it is not industrial agriculture itself that is vulnerable so much as it is the world's food consumers and smallholder farmers. The world's food staples, like corn, soybeans, wheat, and rice, are increasingly treated as commodities for wealth production, not as food. Long bought and sold in the international commodity markets, they are now subject to speculation, just like home mortgages, currency, and gold. Such speculation now drives food prices more strongly than any other single factor (Holt-Gimenez and Patel 2009).

Global Inequality

Despite increases in productivity and yields, hunger persists all over the globe. More than 1 billion people around the world are chronically hungry and more than 870 million are chronically undernourished (CGIAR 2013). With increasing frequency, events such as the spike in global food prices that occurred in 2008 and the major droughts of 2005, 2010, and 2012 create even more hungry people (Bailey 2011). There are also huge disparities in calorie intake and food security between people in developed nations and those in developing nations. At the beginning of the twenty-first century, the world reached a dubious milestone: the number of overweight people (about 1.1 billion) grew roughly equal to the number of underweight people (Gardner and Halweil 2000). This statistic indicates that the unequal distribution of food—which is both a cause and a consequence of global inequality—is at least as serious a problem as the threats to global food production.

Since hunger, poverty, and inequality existed before the rise of industrial agriculture in the latter half of the 1900s, it is tempting to argue that global inequality is unrelated to industrial agriculture—that it has separate causes. While some causes are indeed separate, it is also true that industrial agriculture perpetuates and accentuates existing relationships of inequality. It does this because it is designed to generate profits for the owners of agribusiness concerns and because this process of wealth generation depends on increasing its control of land, farmers, resources, markets, and distribution networks. The inevitable result is the enrichment of some groups and some countries at the expense of others.

Developing nations too often grow food mainly for export to developed nations, using external inputs purchased from the developed nations. While the profits from the sale of the export crops enrich small numbers of elite landowners, many people in the developing nations go hungry. In addition, those with any land are often displaced as the privileged seek more land on which to grow export crops.

Besides causing unnecessary human suffering, relationships of inequality tend to promote agricultural policies and farmer practices that are driven more by economic considerations than by ecological wisdom and long-term thinking. For example, subsistence farmers in developing nations, displaced by large landowners increasing production for export, are often forced to farm marginal lands. The results are deforestation, severe erosion, and serious social and ecological harm. As long as industrial agriculture is based on technology originating in the developed world and on external inputs accessible to so few, the practice of agriculture will perpetuate inequality, and inequality will remain a barrier to sustainability.

PATH TOWARD SUSTAINABILITY

What is the alternative to industrial agriculture? Despite being dedicated to developing forms of sustainable agriculture, the field of agroecology cannot answer this question as directly as the reader might wish. Agroecology consists of principles, concepts, and strategies that must form the foundation of any system of food production that can make a legitimate claim to being a more sustainable successor to industrial agriculture. These principles, concepts, and strategies are more oriented toward offering a design framework for sustainable agroecosystems than they are prescriptions or blueprints for the construction or management of actual agroecosystems, and they do not dictate the specifics of an entire world food system.

Nonetheless, agroecological principles do suggest the general elements of a sustainable food system, and describing these elements will help the reader visualize some of the goals toward which the agroecological approach points.

What Is Sustainability Anyway?

Before describing the elements of a future food system that operates on a more sustainable basis than the industrial agriculture-based system of today, it is helpful to explore what is meant by the term *sustainability.*

As scientists, analysts, activists, and others point with increasing frequency to the *un*sustainability of human society's current systems and practices—everything from fossil-fuel use and industrial agriculture to an economic system dependent on constant growth—it has become ever more common to adopt the label "sustainable." Everyone wants his or her product, industry, alternative method, or proposal to be considered "sustainable." As a result, the term *sustainability* has become increasingly vague, ambiguous, and confusing.

In addition, as a framework for critical analysis of industrial agriculture and for development of alternatives, the concept of sustainability has a key weakness because it depends entirely on an inferred or hypothesized future. Condemning a practice or system as *un*sustainable is essentially to claim that it is bad because it will not last. This sidesteps the possibility that it is causing serious negative consequences right now, in the present. Conversely, arguing for the desirability of a system or practice because it is "sustainable" is really to say that its major benefit would be its durability over time—that we could expect it to still exist at some time in the future. This by itself does not ensure that the system or practice mitigates or reverses harms to people or natural systems. And underlying these drawbacks is a very real practical problem with the concept of sustainability: because sustainability *per se* can never be demonstrated in the present, its proof always remains in the future, out of reach. Thus it is almost impossible to know for sure if a particular practice is in fact sustainable or if a particular set of practices constitutes sustainability.

Despite the drawbacks of the term *sustainability*, however, this text has not abandoned it in favor of another term. In part, that is because there is no good alternative term. Moreover, used precisely and in accordance with its original meaning, *sustainability* really does convey the essence of what we hope to create as an alternative to industrial agriculture—a system of food production, distribution, and consumption that will endure indefinitely because it does not sow the seeds of its own demise. But there is much more to sustainability than mere endurance. As used in this text, *sustainability* refers also to the many characteristics of an ostensibly sustainable practice or system that are responsible for endowing that practice or system with the self-sufficiency, resilience, and balance that *allow it* to endure over time.

If we are going to use the term *sustainable* to indicate the essential feature of what we hope to create as an alternative to industrial agriculture, we should be quite precise about what is entailed in our use of the term. Based on our present knowledge, we can suggest that a "sustainable" food system would, at the very least,

- Have minimal negative effects on the environment and release insignificant amounts of toxic or damaging substances into the atmosphere, surface water, or groundwater;
- Minimize the production of greenhouse gases, work to mitigate climate change by increasing the ability of managed systems to store fixed carbon, and facilitate human adaptation to a warming climate;
- Preserve and rebuild soil fertility, prevent soil erosion, and maintain the soil's ecological health;
- Use water in a way that allows aquifers to be recharged and the water needs of the environment and people to be met;
- Rely mainly on resources within the agroecosystem, including nearby communities, by replacing external inputs with nutrient cycling, better conservation, and an expanded base of ecological knowledge;
- Work to value and conserve biological diversity, both in the wild and in domesticated landscapes;
- Guarantee equality of access to appropriate agricultural practices, knowledge, and technologies and enable local control of agricultural resources;
- Eliminate hunger, ensure food security in culturally appropriate ways, and guarantee every human being a right to adequate food;
- Remove social, economic, and political injustices from food systems.

Each of these features of a sustainable system can be demonstrated in the present, and each one involves undeniable benefits to people and the ecological and social systems on which people depend.

Elements of a Sustainable Food System

Using this list of characteristics of sustainability as a guide, we can envision what food systems of the future might look like—if humankind as a whole begins to follow "the path toward sustainability." Many elements of these systems are already beginning to appear in rough form, alongside industrial food systems, as agroecology grows and spreads.

- The sustainable food system of the future will be made up, in large part, of innumerable small- to medium-scale agroecosystems, each relatively self-contained, adapted to local conditions, and focused primarily on satisfying the food needs of a local population. Only after they satisfy local needs will these agroecosystems attend to the needs of more distant communities.
- Food networks will replace food chains as all players in the food system (from the farm to the table) are reconnected and have a say in what is produced, how it is produced, and how it is exchanged and distributed.
- Traditional, peasant-managed agroecosystems, despite being beleaguered by the encroachment of



FIGURE 1.10 A diverse agroforestry system in the village of Cantagallo, Las Segovias, Nicaragua. A shade-grown organic coffee cash crop below a diverse cover of native and introduced trees, along with other associate crops, provides income, food, firewood, and environmental services such as biodiversity conservation, healthy soil, watershed protection, and carbon sequestration. Food security and more diverse livelihood opportunities provide strong elements of sustainability.

industrial-based systems, still provide more than two-thirds of the world's food. Already embodying many of the key attributes of sustainability, these systems will remain a fundamental basis of food production for much of the world, as their productivity and efficiency are improved through agroecological research.

- Cities—which will continue to provide homes for a large number of the world's people—will be supplied with food less by global markets and more by agroecosystems in the surrounding region and in the cities themselves.
- Agricultural knowledge will exist primarily in the public domain, where it will be widely dispersed and embodied more in farmers' practices than in technological products and systems.
- Farmers will be rewarded for the environmental services that their farms provide beyond the production of food. Protecting biodiversity, producing clean water, stopping soil erosion, sequestering carbon, and promoting the presence of living landscapes will be valued and rewarded.
- Because sustainability in agriculture is not just about growing and raising food, but about how that food is used, distributed, and consumed, a sustainable food system will distribute food more equitably, reduce food overconsumption and waste, and insure that our precious agricultural land is used to feed people rather than automobiles and livestock.
- Food justice will be a common goal in sustainable food systems as food security, food sovereignty, and the right to food become guiding social principles.

It is not an exaggeration to say that the sustainable food system of the future, considered as a whole, will represent a paradigm shift. Like traditional and indigenous agroecosystems, it will conserve resources and minimize exogenous inputs. Like industrial agriculture, it will be very productive. And unlike any system of food production that has heretofore existed on the planet, it will combine these attributes while distributing its benefits equitably among human beings and societies and refraining from displacing its costs onto natural ecosystems increasingly pushed to the brink of collapse. In order for this paradigm shift to come about, agroecology must become a force for change that integrates research, practice, and social change in all parts of our food systems.

Can an Alternative Food System Feed Everyone?

Advocates for industrial agriculture argue that the only way to satisfy the food needs of the expanding world population is to continue to develop new agricultural technologies—particularly GM crop varieties—that will increase yields, reduce insect damage, and eliminate competition from weeds. They dismiss alternative, sustainable, and ecologically based systems as inadequate to the task of growing the needed amount of food. This view is mistaken on at least two accounts.

First, this view exaggerates the need for increasing yields. Globally, the food system currently produces more than enough food calories to adequately feed every single living human being and more (Cassidy et al. 2013). One problem is that 9% of these calories are diverted to make biofuels or other industrial products and another 36% are used for animal feed (less than 10% of which is recovered in the form of animal-based food calories), leaving only 55% to be eaten directly by humans. Another problem is that an estimated *one-third* of the food produced globally is lost to spoilage, spillage, and other problems along the supply chain or simply wasted at the household level (FAO 2013a). In the United States, the amount of food wasted and lost equates to 1249 cal/person/day, which is more than half of what an average person needs (Buzby et al. 2014). Further, the calories that are eaten by humans directly and not lost as waste are distributed very unevenly, with much of them going to expand the waistlines of affluent populations. Thus, the need for more food is driven not as much by the increase in population as it is by wasteful patterns of food use and a shift toward richer diets—*both of which are social choices.* If people ate less animal-based food on average and food was used and distributed more equitably and efficiently, as noted in the previous section, more than enough extra food production capacity would be freed up to feed everyone adequately, leaving a buffer for feeding an expanding population.

Second, this view ignores a growing body of research showing that small-scale, ecologically based, organic, and even traditional peasant systems can approach, match, and even exceed the productivity of industrial systems when measured by the number of people fed per unit of land or the food biomass produced per unit area. These agroecosystems are usually the kinds of diverse, multilayered and integrated systems



FIGURE 1.11 High diversity of production is possible even in the winter on an alternative farm. This farmers' market stand is a direct market option for a 40-acre certified organic family farm located in Santa Cruz County, CA. They grow over 45 varieties of fruits and vegetables, including dry farm tomatoes, shelling beans, and strawberries, and sell their produce to customers and restaurants at 10 farmers' markets in Santa Cruz, Berkeley, and San Francisco.

that will be discussed in Sections IV and V of this text, with a focus on meeting local needs, providing food for the larger communities in which they participate, and maintaining the productive capacity of the soil for the long term. The emphasis of these systems is definitely not on monoculture yield maximization nor the market. A comprehensive 2011 report, presented before the UN Human Rights Council and based on an extensive review of recent scientific literature, showed that agroecologically guided restructuring of agroecosystems has the capability of doubling food production in entire regions within 10 years, while mitigating climate change and alleviating rural poverty (De Schutter 2011).

Role of Agroecology

Many scientists, researchers, and educators in the field of agroecology, and their colleagues in disciplines like agronomy, have long believed that their role is to come up with agricultural methods and systems that are more sustainable, more environmentally friendly, less input dependent, and less technology intensive than those of industrial agriculture. The assumption is that these methods and systems will then be adopted because they are superior when judged by any of various sets of criteria. Unfortunately, the experience of the last couple of decades has exposed the limitations of this view. Although we have accumulated a great deal of knowledge about the ecological relationships underlying sustainable food production, that knowledge has seen relatively little application, and industrial agriculture has meanwhile strengthened its dominance of the world food system.

Transforming agriculture in a fundamental way—putting it on a sustainable path—is going to be a tremendous challenge. A basic assumption of this textbook is that agroecologists can hope to meet this challenge only if we approach it on three different fronts simultaneously.

First, we require more and better knowledge of the ecological relationships among domesticated agricultural species, among these species and the physical environment, and among these species and those of natural systems. This need is satisfied by the **science aspect of agroecology**, which draws on modern ecological knowledge and methods to derive the principles that can be used to design and manage sustainable agroecosystems.

Second, we require effective and innovative agricultural practices, on-the-ground systems that work in the present to satisfy our food needs while laying the groundwork for the more sustainable systems of the future. Satisfying this need is the **practical aspect of agroecology**, which values the local, empirical knowledge of farmers and the sharing of this knowledge, and which undercuts the distinction between the production of knowledge and its application.

Finally, circumstances demand fundamental changes in the ways that humans relate to food, the economic and social systems that determine the distribution of food, and the ways in which food mediates the relationships of power among populations, classes, and countries. Serving this need is the **social-change aspect of agroecology**, which not only advocates for the changes that will lead to food security for all, but also seeks knowledge of the means by which these changes can be activated and sustained.

Although each of these aspects of agroecology is critical, the bulk of this book is dedicated to the science of agroecology. In presenting this material, the book highlights the practical aspect by giving examples of how the science can be successfully applied. The social-change aspect of agroecology is not introduced until Section VI, after the reader has absorbed the full suite of ecological principles and practices that form the foundation of sustainable food systems. The placement of this aspect of agroecology at the end of the book is not an indication of its secondary importance. If agroecologists and others seeking to put agriculture on a more sustainable basis fail to consider the ideas discussed in Section VI, their efforts are likely to be for naught.

FOOD FOR THOUGHT

1. How does the holistic approach of agroecology allow for the integration of the three most important components of sustainability: ecological soundness, economic viability, and social justice?
2. Why has it been so difficult for humans to see that much of the environmental degradation caused by industrial agriculture is a consequence of the lack of an ecological approach to agriculture?
3. What common ground is there between agronomy and ecology with respect to sustainable agriculture?
4. What are the issues of greatest importance that threaten the sustainability of agriculture in the town or region in which you live?
5. What is the meaning of the concept that people "have a right to food"?

INTERNET RESOURCES

Agroecology
www.agroecology.org
A primary site for information, concepts, and case studies in the field of agroecology.

Agroecology in Action
www.agroeco.org
Led by agroecologist Miguel Altieri, Agroecology in Action promotes the integration of agroecological knowledge and technologies into practice while building a deeper understanding of the complex long-term interactions among resources, people, and their environment.

Earth Policy Institute
www.earth-policy.org
Led by the well-known eco-economist Lester Brown, this organization is dedicated to providing a vision of an eco-economy and a roadmap on how to get there. The website provides information on major milestones and setbacks in building a sustainable society.

Food and Agriculture Organization of the United Nations
www.fao.org

Food First: Institute for Food and Development Policy
www.foodfirst.org
Food First is a nonprofit think tank and "education-for-action center" focused on revealing and changing the root causes of hunger and poverty around the world.

Sustainable Table
www.sustainabletable.org
Sustainable Table is a consumer campaign developed by the Global Resource Action Center for the Environment.

Union of Concerned Scientists
www.ucsusa.org
UCS combines independent scientific research and citizen action to develop innovative, practical solutions and to secure responsible changes in government policy, corporate practices, and consumer choices. Its food and agriculture program focuses on the science behind sustainable agriculture as the direction for the future.

Worldwatch Institute
www.worldwatch.org
A nonprofit public policy research organization dedicated to informing policy makers and the public about emerging global problems and trends, and the complex links between the world economy and its environmental support systems. Food and farming are key support systems they monitor.

RECOMMENDED READING

Altieri, M. A. 1995. *Agroecology: The Science of Sustainable Agriculture*, 3rd edn. Westview Press: Boulder, CO.
An important pioneering work on the need for sustainability and a review of the kinds of agroecosystems that will help lead us toward it.

Berry, W. 2009. *Bringing It to the Table: On Farming and Food*. Counterpoint Press: Berkeley, CA.
An eloquent collection of essays by a master farmer and writer that clearly outlines the ethics and culture of human connections to the land and agriculture.

Bohlen, P. J. and G. House (eds.). 2009. *Sustainable Agroecosystem Management: Integrating Ecology, Economics, and Society*. Advances in Agroecology Series. CRC Press/Taylor & Francis Group: Boca Raton, FL.
Through a variety of case studies, this book shows how the agroecosystem concept can be used for developing an interdisciplinary focus for sustainable food system management.

Douglass, G. K. (ed.). 1984. *Agricultural Sustainability in a Changing World Order*. Westview Press: Boulder, CO.
Proceedings of a landmark symposium that helped define the trajectory for future work on the interdisciplinary nature of agricultural sustainability.

Funes-Monzote, F. R. 2008. *Farming Like We're Here to Stay: The Mixed Farming Alternative for Cuba*. Wageningen University: Wageningen, the Netherlands.
A motivational account of the remarkable effort to develop food self-sufficiency as a response to economic crisis in Cuba, and the grounding of this response in the concepts of agroecology.

Gliessman, S. R. and M. E. Rosemeyer. 2010. *The Conversion to Sustainable Agriculture: Principles, Processes, and Practices*. Advances in Agroecology Series. CRC Press/Taylor & Francis Group: Boca Raton, FL.
The framework for the conversion of food systems to sustainability.

Goodman, D., M. DuPuis, and M. K. Goodman. 2011. *Alternative Food Networks: Knowledge, Place and Politics*. Routledge: London, U.K.
A critical review of the growth of alternative food networks and their struggle to defend their ethical and aesthetic values against the standardizing pressures of the corporate mainstream with its "placeless and nameless" global supply networks.

Guzmán-Casado, G., M. Gonzáz de Molina, and E. Sevilla-Guzmán. 1999. *Introducción a la Agroecología como Desarrollo Rural Sostenible.* Ediciones Mundi-Prensa: Madrid, Spain. (In Spanish).
A pioneering description of agroecology as a social movement focused on sustainable rural development, with a strong emphasis on the European model.

Halweil, B. 2004. *Eat Here: Reclaiming Homegrown Pleasures in a Global Supermarket*. Worldwatch Institute: Washington, DC.
An engaging analysis of the current crisis in farm and food systems, accompanied by a convincing argument for reconnecting what we eat with how and where food is grown.

International Assessment of Agricultural Knowledge, Science and Technology for Development. 2009. Agriculture at a crossroads. Global report by the International Assessment of Agricultural Knowledge, Science and Technology for Development (IAASTD). Island Press: Washington, DC.
The final report of a broad consortium of international agencies and experts that presents an alternative and promising approach for resolving the interrelated global problems of hunger, rural poverty, and sustainable development.

Jackson, W., W. Berry, and B. Colman (eds.). 1986. *Meeting the Expectation of the Land*. Northpoint Press: Berkeley, CA.
A collection of contributions from a diverse set of experts, designed to inform the general public of the people- and culture-based elements that are needed to make the transition to a sustainable agriculture.